Project Title

SwiftOps: Smart Operations & Decision Intelligence System

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1. Project Overview

Modern delivery-based companies (like food delivery, e-commerce, logistics platforms) face challenges in managing operations efficiently. Issues like delayed deliveries, stock imbalance, high operational costs, and poor decision-making affect business performance.

This project aims to build a **data-driven operations intelligence system** using **Python, SQL, and Power BI**, enabling businesses to monitor, analyze, and optimize operations from a single dashboard.

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2. Project Objectives

1️⃣ **Optimize Delivery Performance**

- Identify delays in shipping and processing
- Improve delivery efficiency

2️⃣ **Improve Inventory Management**

- Balance stock across warehouses
- Reduce stockout and overstock issues

3️⃣ **Demand Forecasting**

- Analyze historical sales trends
- Predict future demand

4️⃣ **Cost Optimization**

- Reduce operational, storage, and logistics costs
- Improve profit margins

5️⃣ **Decision Intelligence**

- Build a system for real-time insights
- Enable business owners to take quick decisions

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3. Feature Engineering (Mandatory)

Intern must create the following new features:

- Inventory_Turnover = Monthly_Sales / Current_Stock
- Delivery_Efficiency = Shipping_Time_Days + Order_Processing_Time

- Total_Cost = Operational_Cost + Storage_Cost + Transportation_Cost
- Estimated_Profit = Monthly_Sales - Total_Cost
- Stock_Utilization = Current_Stock / Warehouse_Capacity
- Demand_Gap = Demand_Forecast - Current_Stock
- Risk_Score = Stockout_Risk + Return_Rate + Damaged_Goods

These features are critical for advanced analysis.

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4. Data Cleaning & Preprocessing

Using Python (pandas):

- Handle missing values
- Remove duplicates
- Fix incorrect data types
- Detect and treat outliers

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5. Python Tasks

Intern must:

- Load dataset using pandas
- Perform EDA (Exploratory Data Analysis)
- Apply feature engineering
- Create visualizations:
    - Sales trends
    - Cost analysis
    - Delivery performance
    - Inventory distribution

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6. SQL Analysis Tasks

Intern must write queries for:

- Top delayed warehouses
- High stockout risk locations
- Product-wise sales performance
- Cost vs profit analysis

- Demand vs stock comparison

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7. Power BI Dashboard (Main Deliverable)

Dashboard Requirements:

KPI Cards:

- Total Sales
- Total Profit
- Avg Delivery Time
- Stockout Risk

Visualizations:

- Warehouse performance (map/chart)
- Sales vs Cost comparison
- Demand forecast trend
- Inventory vs Capacity
- Delivery time analysis

Filters:

- Location
- Product Category
- Warehouse ID
- Supplier

Goal:
Business owner should manage entire operations from one dashboard

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8. Expected Outcome

After project completion:

- ✓ Reduced delivery delays
- ✓ Balanced inventory
- ✓ Better demand forecasting

- ✓ Improved profitability
- ✓ Real-time monitoring system

9. Final Deliverables

Intern must submit:

- ✓ Cleaned dataset
- ✓ Feature engineered dataset
- ✓ Python notebook (.ipynb)
- ✓ SQL queries file
- ✓ Power BI dashboard (.pbix)
- ✓ Final project report (PDF)